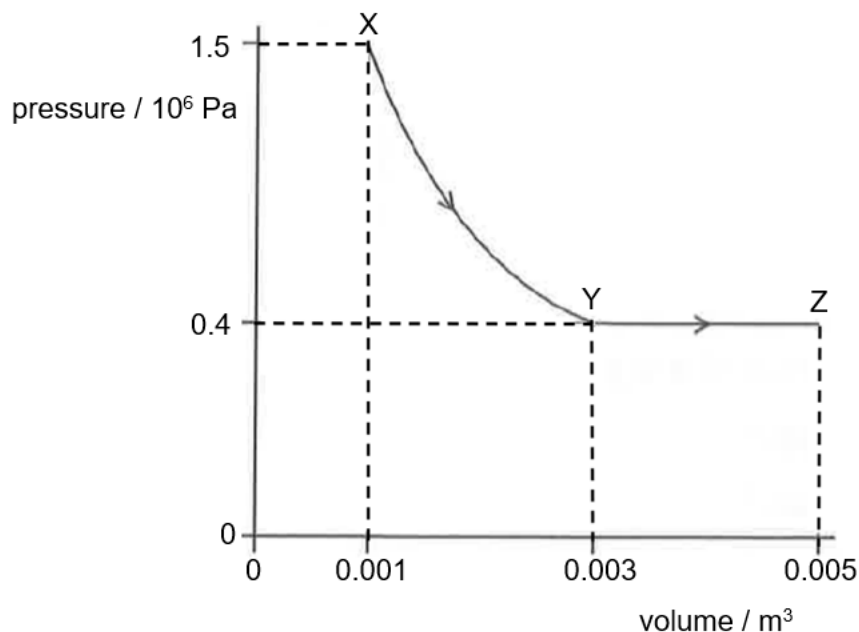
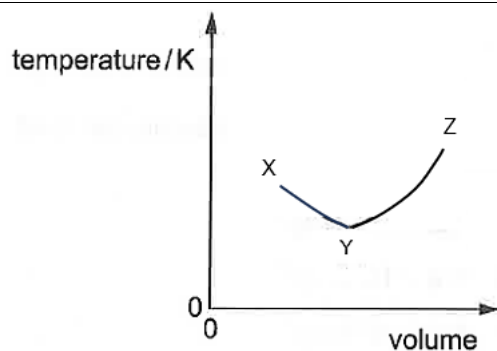


13 A fixed mass of ideal gas undergoes changes of pressure and volume starting at X, as shown.

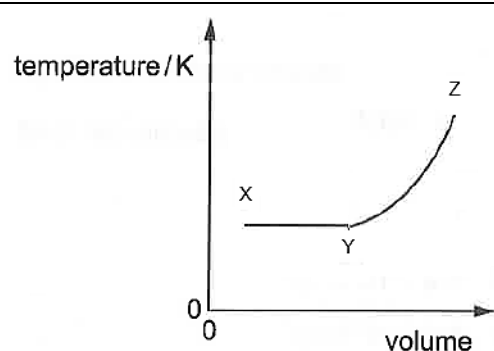


Which graph shows how temperature (measured in kelvin) changes with volume?

A



B



C

D

		temperature/K X Y Z 0 0 volume	temperature/K X Y Z 0 0 volume
L2	Answer: A Apply $pV = nRT$ At X, $(1.5 \times 10^6 \times 0.001) = nRT$ $T = \frac{1.5 \times 10^3}{nR}$ At Y, $(0.4 \times 10^6 \times 0.003) = nRT$ $T = \frac{1.2 \times 10^3}{nR}$ At Z, $(0.4 \times 10^6 \times 0.005) = nRT$ $T = \frac{2.0 \times 10^3}{nR}$ temperature $T_Z > T_X > T_Y$		